

Arbeitsblatt: Lineare Funktionen — Klasse 8

A linear function has the form $y = mx + b$. The number m is called the slope and the number b is called the y-intercept.

1. The slope

The slope m tells you how steep the line is. It is computed as the rise divided by the run: $m = \frac{\Delta y}{\Delta x}$.

2. The y-intercept

The number b is the value of y where the line crosses the vertical axis, that is, when $x = 0$.

3. From two points to an equation

Given two points, first compute m using the rise over run, then substitute one point into $y = mx + b$ and solve for b .

Common student error: many students believe a steeper line always has a larger value of b , confusing steepness with height.